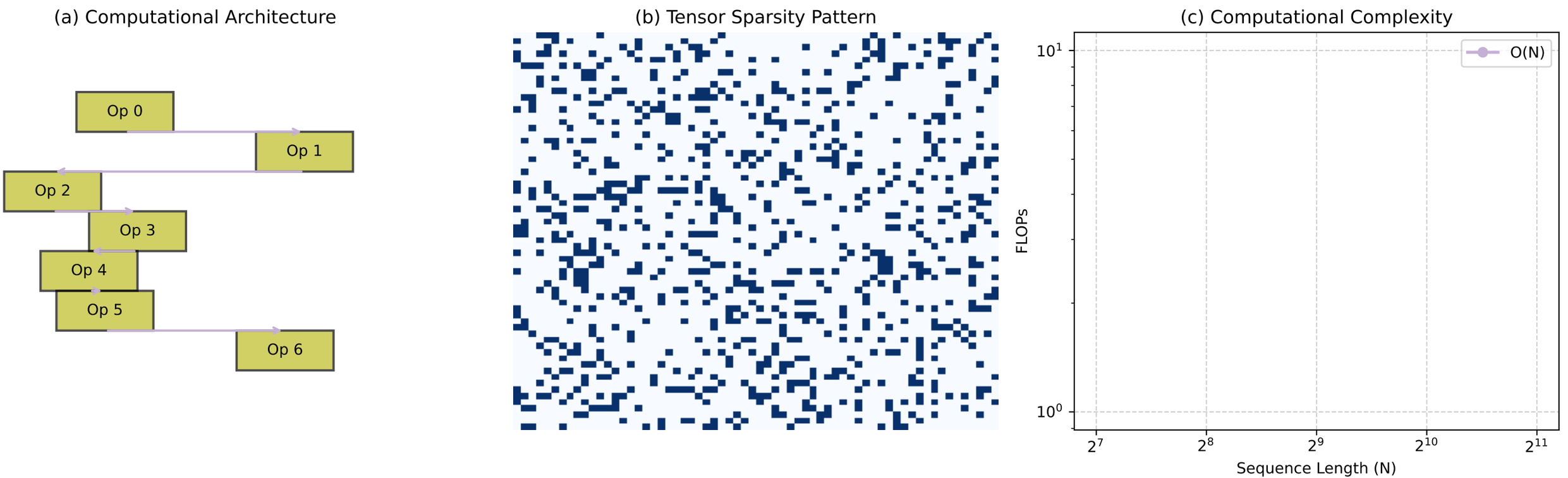Module Architecture: two_phase_benchmark